Decoding the MoLF Experts

Grounding mixture-of-experts routing in pathology concepts
through a vision–language embedding space

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Abstract

MoLF predicts spatial gene expression from H&E histology with a flow-matching model whose velocity field is a mixture of experts. The experts specialise, but what each of them responds to cannot be read from the model. We retrained MoLF on image features from CONCH, a pathology vision–language model, so that the router operates on embeddings that share a space with text, and assembled a vocabulary of 2,450 visual pathology concepts from a pathologist-curated bank, biomedical ontologies and PubMed. On 23 held-out HEST-1k slides the retrained model reaches a mean slide Pearson correlation of 0.246 and a masked MSE of 0.370. Both depend strongly on the spatial transcriptomics platform: Xenium slides reach a correlation of 0.46 against 0.15 for Visium, yet account for 38% of the squared error while holding 6% of the measured entries. Routing is highly non-uniform, and one expert is used almost exclusively on Xenium slides. Scoring each expert's highest-affinity patches against the vocabulary gives recognisable themes (inflammation, lymphoid structures, glandular and mucinous epithelium), but an expert's concept profile correlates only weakly between slides (mean rank correlation 0.07–0.24 per expert), and for every expert the best-agreeing slide pairs come from the same data source and platform. So besides morphology the experts also pick up the measurement platform, which acts as a batch effect here, and the data do not support giving each expert a single pathology label.

1 Introduction

Spatial transcriptomics measures gene expression at known positions in a tissue section [7, 22]. Because the same section is usually stained with haematoxylin and eosin (H&E), paired data sets such as HEST-1k [8] make it possible to learn a map from routine histology to local expression [4]. MoLF [6] approaches this as generative modelling: a conditional flow [14] transports Gaussian noise to the latent space of a gene autoencoder, and the velocity field is a mixture of experts [21] in which a router assigns every tissue patch to two of six experts. The original work reports state-of-the-art accuracy on the HEST-1k pan-cancer benchmark and shows that the experts specialise. It does not say what they specialise in.

This project asks whether the experts can be described in pathology terms. Concept-based interpretability usually needs labelled examples of each concept [1, 9]. Vision–language models remove that requirement by embedding images and free text in one space [20]; CONCH [16] does this for pathology. If the model's own input is a CONCH image embedding, an expert's preferred patches can be compared with any written concept without further training. The original MoLF uses UNI-v2 features [2], which have no text counterpart, so the model had to be retrained.

In this project we (i) retrained MoLF on frozen CONCH patch embeddings with a frozen 1,386-gene variational autoencoder and evaluated it on held-out slides, with the inference protocol fixed on validation data; (ii) built a concept vocabulary from a pathologist-curated reference bank, ontologies and PubMed, and audited its failure modes; (iii) scored each expert against this vocabulary on 23 test slides, keeping router affinity and actual routing apart and measuring how

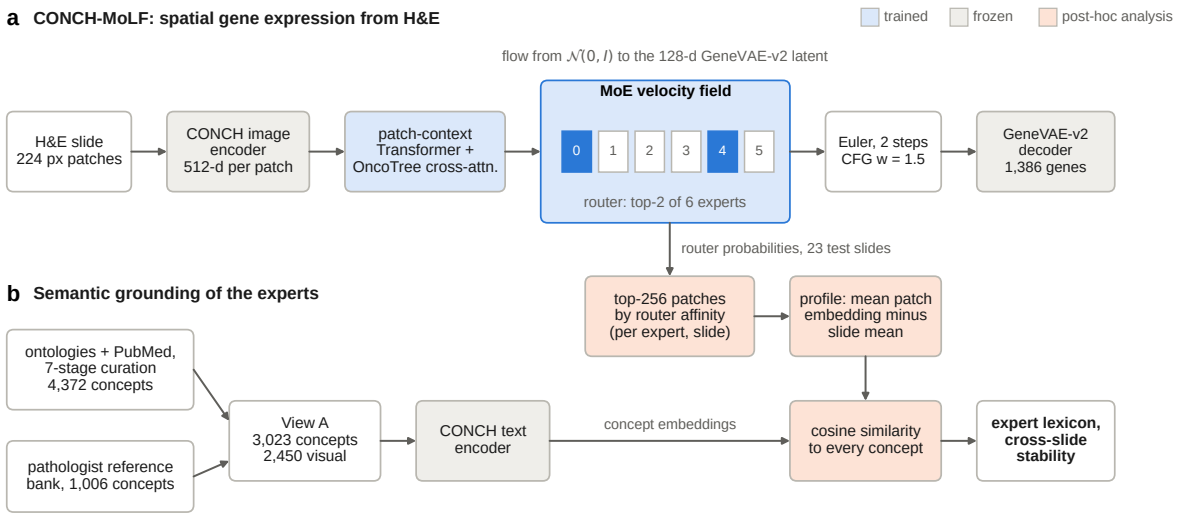


Figure 1. Overview. **a**, CONCH-MoLF. Frozen CONCH embeddings of H&E patches pass through a patch-context Transformer with OncoTree cross-attention; a top-2 router mixes six experts into a velocity field in the latent space of a frozen GeneVAE-v2, which decodes the sampled latent into 1,386 genes. **b**, Interpretation. Concepts from a pathologist-curated bank and from ontologies and PubMed are embedded with the CONCH text encoder; each expert's 256 highest-affinity patches on a slide form a contrastive profile that is scored against every concept.

stable the result is across slides and platforms; and (iv) found that the platform, and not only morphology, shapes both the prediction error and the routing. Fig. 1 gives an overview of the pipeline.

2 Methods

2.1 Data

We used 504 HEST-1k slides [8] split into 385 training, 96 validation and 23 test slides (Fig. 2a). The platforms are Visium (279/69/16 slides in the three splits), Xenium (42/9/7), legacy Spatial Transcriptomics (59/17/0) and Visium HD (5/1/0). Each slide carries an OncoTree cancer-type code [11]; the conditioning vocabulary has 29 codes, of which the test set covers 10 (Fig. 2b). Slides do not overlap between splits, but four of the seven Xenium test slides share a patient or study with training or validation slides, so results describe unseen slides rather than unseen patients.

The targets are 1,386 genes, 1,169 from the MSigDB Hallmark collection [13] and 217 highly variable genes, as in upstream MoLF, with $\log(1 + x)$-transformed counts. Genes not measured on a slide are masked in every loss and metric; this matters for Xenium, whose targeted panels cover only a small part of the gene list (Section 3.2). The image inputs are 224×224-pixel H&E patches centred on the measured spots, embedded once with the frozen CONCH v1 image encoder into 512-dimensional, ℓ_2-normalised vectors (1,076,882 patches in total). Spot coordinates are not model inputs.

2.2 Model

Target space. A GeneVAE-v2 [10] with a Transformer encoder maps the expression vector $y_i \in \mathbb{R}^{1386}$ of patch i to a 128-dimensional latent posterior and decodes latents back to expression. It was trained separately (best epoch 929) and is frozen in all experiments; the flow is trained on posterior samples $z_1 \sim q_\phi(z \mid y)$.

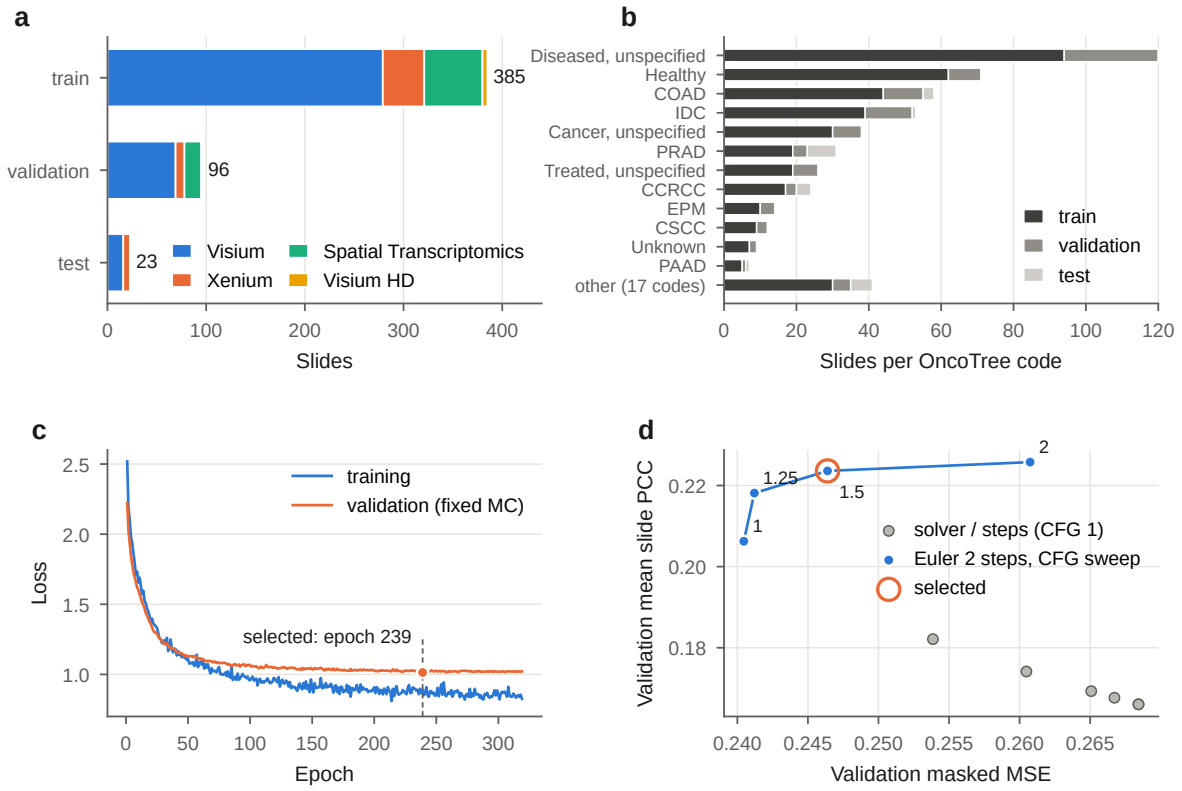


Figure 2. Data, training and inference protocol. **a**, Slides per split and platform. **b**, Slides per OncoTree code (the twelve most frequent codes; the remaining 17 are pooled). **c**, Training and validation loss (flow matching plus gene reconstruction); the validation loss uses fixed Monte-Carlo noise so that epochs are comparable. Epoch 239 was selected and training stopped at epoch 319. **d**, Inference protocols on the 96 validation slides. Fewer Euler steps lowered the error; guidance traded error for correlation, and guidance 1.5 was chosen between the two extremes.

Conditional flow. The flow uses the conditional optimal-transport path of Lipman et al. [14]: with $z_0 \sim \mathcal{N}(0, I)$ and $t \sim U[0,1]$, $z_t = (1-t)\, z_0 + t\, z_1$ and the target velocity is $z_1 - z_0$. The velocity network $v_\theta(z_t, t; x, c)$ sees the latent state, the time, the CONCH embeddings x of all patches of the slide and the OncoTree code c. A one-layer Transformer without positional encoding mixes information between the patches of a slide, so the model is equivariant to patch order, and the OncoTree code enters through cross-attention.

Mixture of experts. Each of the six experts is a one-layer Transformer block (width 256, four heads). The input h_i of the router and the experts concatenates, for patch i, the latent state z_t, the time t and the context-aware, OncoTree-conditioned image embedding, so routing depends both on the image and on the state of the flow. The router produces logits $g(h_i) \in \mathbb{R}^6$. We call $\pi_e(i) = \mathrm{softmax}(g(h_i))_e$ the *affinity* of patch i for expert e. Only the two experts with the largest logits, K_i, are evaluated, with *route weights*

$$w_e(i) = \frac{\exp g_e(h_i)}{\sum_{k \in K_i} \exp g_k(h_i)} \text{ for } e \in K_i, \qquad w_e(i) = 0 \text{ otherwise}, \tag{1}$$

and the velocity is $\sum_e w_e(i)\, E_e(h_i)$. Affinity and route weight answer different questions, and Section 3.3 shows that they can disagree strongly.

Objective. The loss is

$$\mathcal{L} = \left\| v_\theta(z_t, t; x, c) - (z_1 - z_0) \right\|^2 + \mathcal{L}_{\text{gene}} + 0.01\, \mathcal{L}_{\text{bal}}, \tag{2}$$

where $\mathcal{L}_{\text{gene}}$ is the masked squared error between y and the decoded one-step estimate $\hat{z}_1 = z_t + (1-t)\, v_\theta$, and $\mathcal{L}_{\text{bal}} = 6 \sum_e f_e P_e$ is the load-balancing term of Fedus et al. [3], with f_e the mean affinity and P_e the fraction of top-2 assignments to expert e. The balancing term is applied during training only.

Guidance. For classifier-free guidance [5] the image embeddings are replaced by a learned null embedding with probability 0.1 during training. At inference the guided velocity is $v_\varnothing + \omega\,(v_x - v_\varnothing)$.

2.3 Training and inference protocol

The model was trained with AdamW (learning rate 5×10^{-5}, weight decay 0.01, cosine schedule with 5% warm-up), one slide per batch, for at most 400 epochs. Early stopping monitored a validation loss computed with five fixed noise draws (patience 80 epochs, minimum improvement 10^{-4}). The best epoch was 239; training stopped at epoch 319 (Fig. 2c).

The inference protocol was selected on the validation slides only (Fig. 2d). A first stage compared Euler (10, 25, 50 steps) and RK4 (10, 25 steps) at guidance 1.0; error fell monotonically as the number of steps decreased, and a second stage confirmed that two Euler steps were best. A third stage varied the guidance scale (1.0, 1.25, 1.5, 2.0) and a fourth repeated the three main candidates with four more noise seeds. Guidance 1.0 gave the lowest error and 2.0 the highest correlation; we fixed Euler, two steps and guidance $\omega = 1.5$. The test slides were evaluated once, with this protocol and five noise seeds, after all choices had been made.

Metrics. Errors (MSE, RMSE, MAE) are pooled over all measured entries. For correlation we compute, for every slide and gene, the Pearson correlation across the slide's spots. The *slide PCC* of a slide is the mean over its genes; the *mean gene PCC* averages over slides first and then over genes. As a reference we also report how well GeneVAE-v2 reconstructs the measured expression. This is a practical ceiling for any model that predicts the latent state.

2.4 Concept vocabulary

Sources. The *reference bank* contains 1,006 concepts curated by a pathologist in five tiers: morphology, cell type, niche, gene programme and pathway. The *harvested bank* was collected automatically from NCIt, SNOMED CT, the Cell Ontology and GO (through BioPortal), MSigDB gene sets and PubMed abstracts, the latter with scispaCy entity recognition [18]; UBERON supplied a list of organ names used during curation. After curation it contains 4,372 concepts, most of them from the Cell Ontology (1,474), NCIt (1,211) and SNOMED CT (1,082), with 304 MSigDB gene sets, 283 PubMed terms and 18 GO terms.

Curation. The harvested candidates pass through seven curation stages (Table 1). All comparisons between labels use SapBERT [15], a model trained to embed synonymous biomedical names close together. Stages 2 to 4 decide which candidates are kept, and stages 5 to 7 organise the survivors and prepare their prompts. The filter and the tier refinement compare candidates with the reference bank, so the reference vocabulary acts as the definition of a relevant concept. The thresholds are the same for all tiers: a per-tier calibration was planned, but it left every tier at the default values.

View A. Matching the two banks with SapBERT (similarity at least 0.85 within a tier, 0.92 across tiers) gives a merged bank of 5,378 concepts. The subset used for interpretation, View A, has 3,023 concepts (Fig. 7a): all 1,006 reference concepts, the 255 harvested concepts that matched a reference concept of the same tier, and 1,762 further harvested concepts whose label passes simple checks (at least 4 characters, at most 8 words) and lies within a similarity of 0.70 of a reference concept of the same tier. The other 2,355 harvested concepts are left out.

Table 1. The seven curation stages applied to the harvested candidates. Similarities are cosine similarities between SapBERT embeddings of the concept labels.

	Stage	What it does
1	Embed	Embeds every label with SapBERT.
2	Deduplicate	Clusters labels within a tier (HDBSCAN [17], similarity ≥ 0.85; gene sets also by gene overlap ≥ 0.5) and keeps one label per cluster, preferring SNOMED CT, then NCIt, the Cell Ontology, GO, MSigDB and PubMed.
3	Filter	Compares each label with the mean embedding of the reference concepts of its tier. Drops it below 0.35, or if fewer than half of its 10 nearest reference concepts share its tier, and keeps it above 0.55; labels in between are marked ambiguous. Labels closer to a negative class (diseases, anatomical sites, procedures, clinical metadata) than to their own tier can also be dropped.
4	Adjudicate	A local Qwen2.5-3B model [19] decides keep or drop for the ambiguous labels, and here also for the kept ones, following a written rubric: keep visible features, cell types, niches, programmes and pathways; drop diseases, anatomical sites, procedures, clinical fields and generic terms.
5	Refine tiers	Moves a concept to another tier when at least two of its three nearest reference concepts belong there (nearest similarity ≥ 0.5), and adds organ and category information.
6	Canonicalise	Assigns concept identifiers, writes labels in sentence case and removes invalid records.
7	Write prompts	Generates a human and a mouse prompt for each concept from tier-specific templates.

Before embedding, 115 same-tier duplicates were collapsed and the tiers were split into a visual set (morphology, cell type, niche; 2,450 concepts) and a transcriptomic set (gene programme, pathway; 458 concepts) whose content is not visible on H&E. Each concept has a prompt built from a tier-specific template, for example *"Histopathological feature: comedo necrosis observed in H&E stained human tissue"*, embedded with the CONCH text encoder. The analysis uses the 2,450 visual concepts: 1,243 morphology, 1,123 cell type and 84 niche.

2.5 Expert–concept alignment

Each test slide was run through the frozen model with the frozen protocol and five noise seeds. At both Euler states we recorded, from the conditional branch, the affinity $\pi_e(i)$ and route weight $w_e(i)$ of every patch; the per-patch values are averaged over seeds and states. For slide s and expert e, the 256 patches with the highest affinity form the set $T_{e,s}$. Their contrastive profile is

$$m_{e,s} = \frac{1}{|T_{e,s}|} \sum_{i \in T_{e,s}} x_i - \frac{1}{|S_s|} \sum_{i \in S_s} x_i, \tag{3}$$

where S_s is the set of all patches on the slide. Subtracting the slide mean removes what all patches of a slide share (tissue type, stain, scanner) and keeps what distinguishes the expert's patches. Each concept k with text embedding u_k receives the score $\cos(m_{e,s}, u_k)$. The *expert lexicon* ranks concepts by their mean score over the 23 slides; for each concept we also count the slides on which it is among the expert's ten best.

Stability. For every expert and every pair of test slides we computed the Spearman correlation between the two slides' score vectors over all concepts. The 253 pairs split into 120 Visium–Visium, 21 Xenium–Xenium and 112 cross-platform pairs.

Filtered analysis. A review of the scored concepts found 188 that cannot be H&E findings: 48 Cell Ontology cell types of other species, one cytology-only finding and 139 cell subsets defined only by molecular markers or functional state (Fig. 7b). A concept's score does not depend

Table 2. Gene-expression prediction with the selected protocol (Euler, two steps, guidance 1.5). Mean ± SD over five noise seeds; arrows mark the better direction.

Metric	Validation (96 slides)	Test (23 slides)
Masked MSE ↓	0.2464 ± 0.0001	0.3699 ± 0.0002
Masked RMSE ↓	0.4964 ± 0.0001	0.6082 ± 0.0002
Masked MAE ↓	0.2974 ± 0.0000	0.3677 ± 0.0001
Mean slide PCC ↑	0.2236 ± 0.0006	0.2464 ± 0.0004
Median slide PCC ↑	0.1599 ± 0.0038	0.1846 ± 0.0016
Mean gene PCC ↑	0.1937 ± 0.0005	0.1662 ± 0.0005

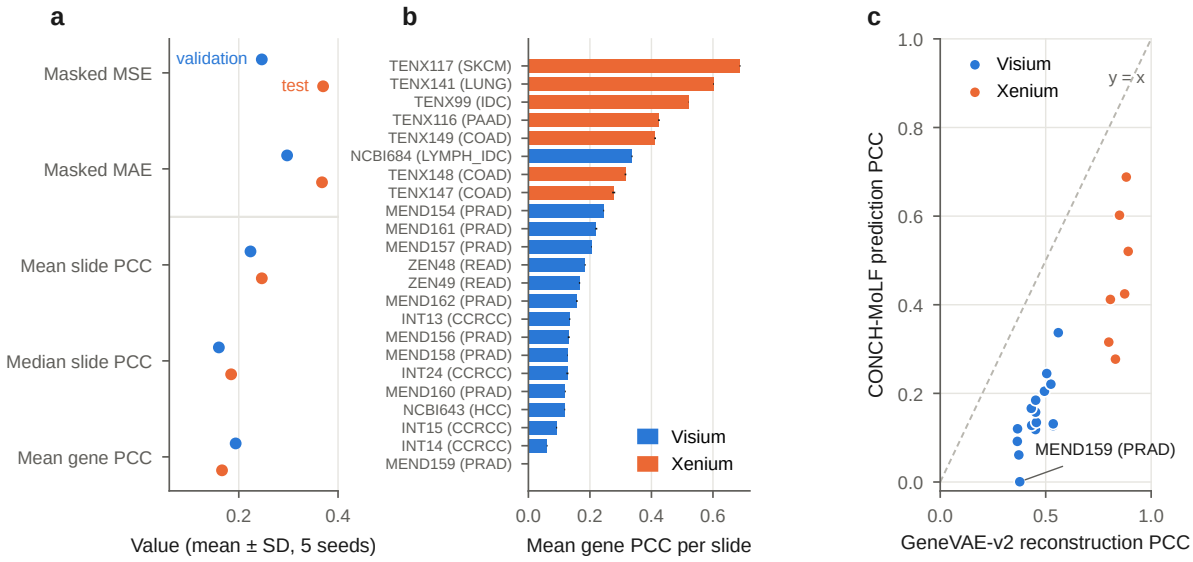


Figure 3. Prediction on held-out slides. **a**, Validation and test metrics. **b**, Slide PCC for each test slide (OncoTree code in brackets), mean ± SD over five seeds, coloured by platform. **c**, Slide PCC of the model against that of the GeneVAE-v2 reconstruction of the same slide; points on the diagonal would reach the reconstruction ceiling.

on the other concepts, so we recomputed the lexicon and the stability without them from the per-slide scores. As a check, the recomputation reproduces the original aggregate when nothing is removed. All lexicon and stability results below are the filtered ones.

3 Results

3.1 Gene-expression prediction

Table 2 lists the validation and test results. The standard deviation over the five noise seeds is below 0.004 for every metric, so sampling noise is negligible at this protocol. The test slides show higher correlation but also higher error than the validation slides (Fig. 3a). This comes from the mix of platforms: 7 of the 23 test slides are Xenium, against 9 of 96 in validation, and Xenium slides are predicted with higher correlation and larger absolute error. Within each platform the test correlation is in fact lower than on validation (Visium 0.15 against 0.20, Xenium 0.46 against 0.53).

Per slide, correlation ranges from 0.001 to 0.69 (Fig. 3b). The failure case is MEND159, a prostate adenocarcinoma: its error equals the test median, but its predicted spatial pattern is uncorrelated with the measured one, although the autoencoder reconstructs the slide with a slide PCC of 0.38.

Table 3. Test results by platform. Slide PCC is averaged over slides and MSE is pooled over entries. GeneVAE-v2 is the reconstruction of the measured expression by the frozen autoencoder. The last column is each platform's share of the total squared error.

		Mean slide PCC		Masked MSE		Share of
Platform	Slides	MoLF	GeneVAE-v2	MoLF	GeneVAE-v2	squared error
Visium	16	0.152	0.457	0.244	0.116	62%
Xenium	7	0.463	0.848	2.193	0.440	38%

3.2 Platform effects

The two platforms produce very different targets. A Xenium test slide measures on average 155 of the 1,386 genes, against 1,376 for Visium, and its measured values are much larger (mean 1.93 against 0.31 after the log transform) and less often zero (22% against 72%). Prediction quality differs accordingly (Table 3). Xenium slides are predicted with a slide PCC of 0.46 against 0.15 for Visium, and the 7 Xenium slides contribute 38% of the total squared error while holding 6% of the measured entries. Relative to the autoencoder ceiling, the model's error is 2.1 times the reconstruction error on Visium and 5.0 times on Xenium: the autoencoder represents both platforms well (slide PCC 0.46 and 0.85), and most of the remaining error arises when the latent state is predicted from the image (Fig. 3c). Four of the seven Xenium test slides have related specimens in the training data; they are not the best-predicted ones (slide PCC 0.28–0.52, against 0.42–0.69 for the other three).

In this data set, platform is confounded with cancer type and study. Of the 16 Visium test slides, 8 are prostate and 4 renal carcinomas, while the 7 Xenium slides come from pancreatic, skin, lung, colon (three sections) and breast cancer. The platform difference is therefore not a clean measurement of the technology alone.

3.3 Expert routing

Across the 92,507 test patches, routing is far from uniform (Fig. 4a). Slide-averaged route weights are 0.240, 0.025, 0.128, 0.007, 0.338 and 0.262 for experts 0 to 5. Experts 4, 5 and 0 carry most of the computation. Expert 3 is almost never used: its route weight stays below 0.04 on every slide. Expert 1 is used only rarely. Routing also depends on platform (Fig. 4b). Expert 2 takes 0.38 of the route weight on Xenium slides and 0.02 on Visium, where experts 4 and 5 dominate.

Affinity and use can diverge. Selecting patches by affinity is the natural way to find what an expert "prefers", but for Expert 3 the 256 highest-affinity patches are routed to it only 8% of the time, while for Expert 4 the figure is 83% (Fig. 4c). For rarely used experts the selected patches therefore describe a preference the model seldom acts on. We report all six experts, but the profiles of experts 1 and 3 say little about the model's computation.

3.4 Expert lexicon

Fig. 5 shows the six best concepts per expert. Each list has a recognisable theme. Expert 0 aligns with lymphoid and follicular structures (lymphocytic infiltrate, marginal-zone B cells, tertiary lymphoid structures) and with cribriform and intraductal gland patterns. Experts 1, 2 and 3 align with inflammation of different kinds, and Expert 2 also with vessel walls and serosal surfaces. Expert 4 aligns with glandular and mucinous epithelium (mucinous neoplasms, goblet cells, villous architecture) and Expert 5 with inflamed stroma and smooth muscle. Inflammation terms dominate experts 1 to 3 and also appear for Expert 5. The vocabulary contains 43 morphology concepts named "... inflammation" that differ only by a qualifier (acute, diffuse, multifocal, necrotising), so near-synonyms can fill several places of one list; all six of Expert 1's concepts are of this kind.

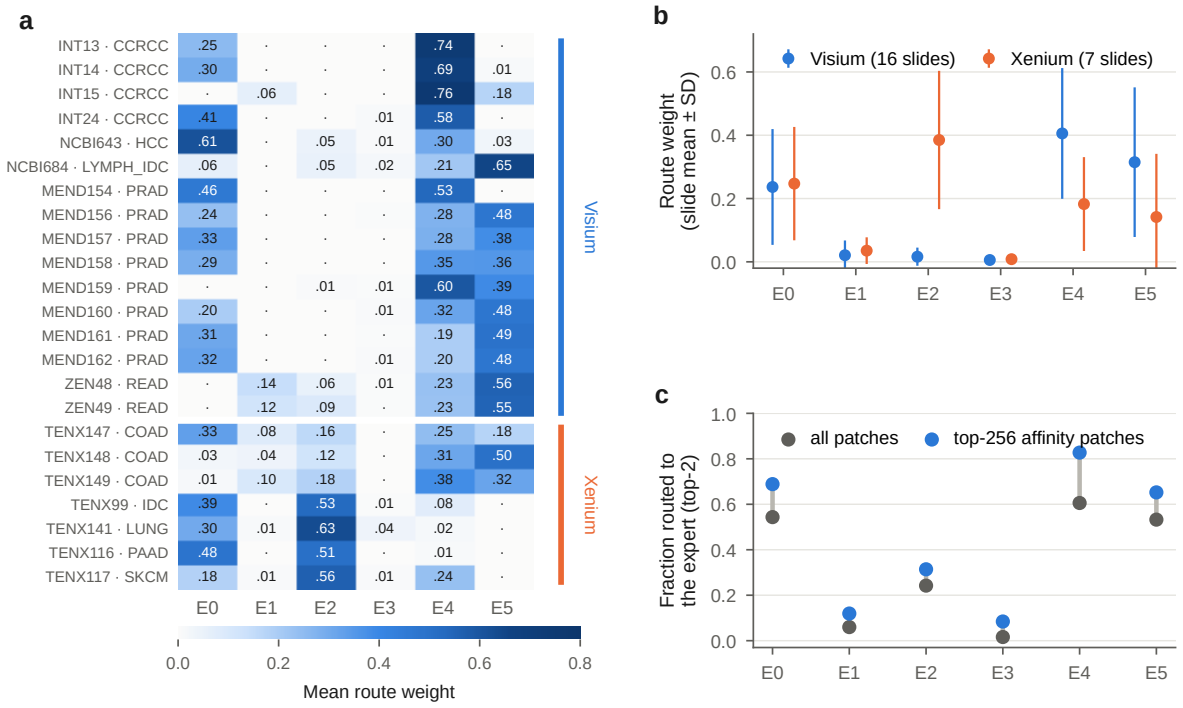


Figure 4. Natural routing on the 23 test slides. **a**, Mean route weight per slide and expert, slides grouped by platform and cancer type. **b**, Route weight by platform (mean ± SD over slides). **c**, Fraction of time an expert is among the two selected experts, over all patches and over its 256 highest-affinity patches per slide.

The lexicon is an average, and the numbers next to each concept show how little of it holds on an individual slide. No concept is among an expert's ten best on more than 8 of the 23 slides, and three of the 36 concepts shown are in the top ten on no slide at all. They rank high on average because they score reasonably well on many slides without being among the best ten on any of them.

3.5 Stability across slides and platforms

The per-slide concept profiles of an expert are only weakly related (Fig. 6). The mean Spearman correlation between two slides' profiles ranges from 0.07 (Expert 0) to 0.24 (Expert 3), and individual pairs span from strongly negative to strongly positive. For every expert the highest mean correlation is found within one platform, and for experts 1, 2 and 5 the cross-platform correlation is close to zero. The pairs that agree best share more than the platform: for every expert, all 10 most similar pairs come from the same HEST data source (the prefix of the sample identifier, such as MEND for the eight prostate slides), although such pairs are only 57 of the 253. So an expert's patches look different from slide to slide, and where the profiles do agree, the slides come from the same cohort, which shares tissue type, preparation and platform. Removing the invalid concepts changes the mean correlations by at most 0.014.

3.6 Quality of the concept vocabulary

Reviewing the vocabulary (Fig. 7) showed where the curation falls short. The filter in stage 3 compares concept names with the reference vocabulary, so it measures how much a name resembles pathology terminology rather than whether the concept can be seen on H&E, and the curation has no species filter. All 48 cross-species entries came from the Cell Ontology, which covers all animals; examples are insect fat-body and eggshell-secreting cells, which neither the

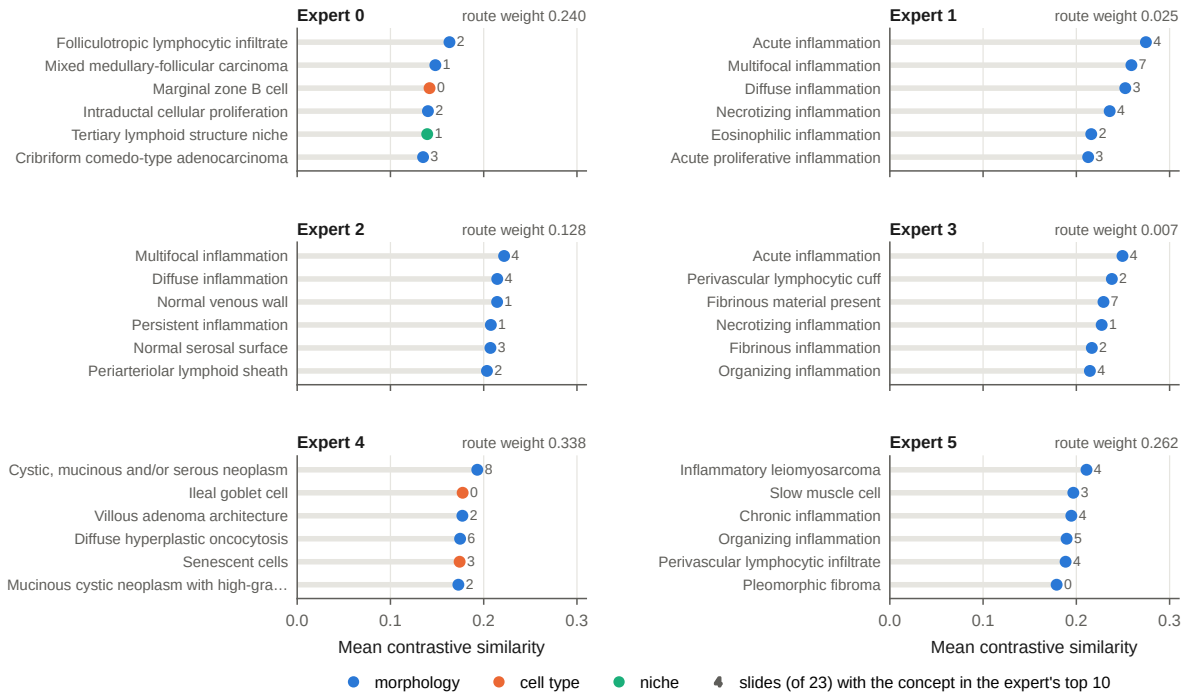


Figure 5. Expert lexicon. The six concepts with the highest mean contrastive similarity for each expert, averaged over 23 test slides, after excluding the 188 invalid concepts. Colour gives the concept tier; the number next to each point is the number of slides on which the concept is among the expert's ten best. The route weight is the slide-averaged share of computation the expert receives.

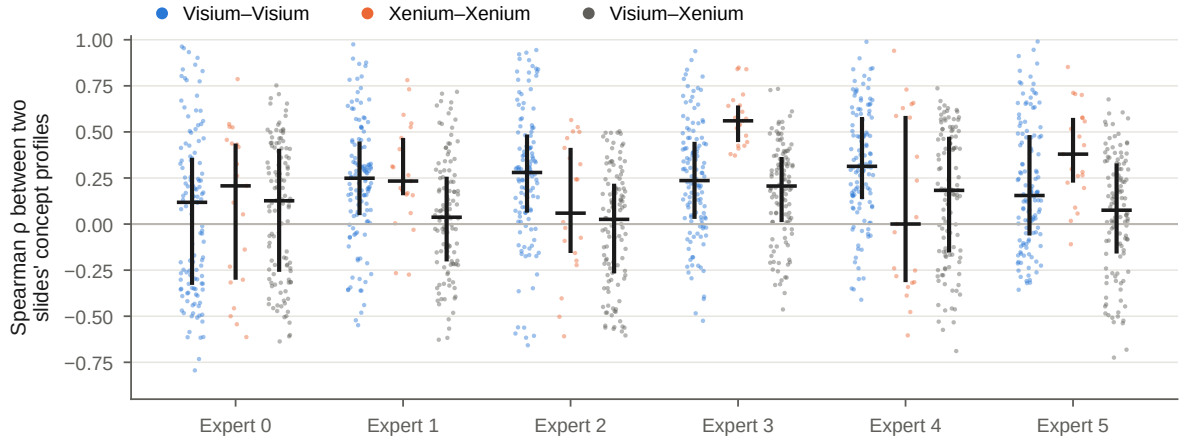


Figure 6. Cross-slide stability of the concept profiles. Each point is one pair of test slides; its value is the Spearman correlation between the two slides' scores over all retained concepts. Bars show the interquartile range and median for Visium–Visium (120 pairs), Xenium–Xenium (21) and cross-platform pairs (112).

filter nor the LLM step removed. Marker-defined subsets such as memory T-cell or regulatory T-cell populations look like their parent cell type on H&E, and 24 of them came from the reference bank itself. Six of the 60 top-ten entries of the original, unfiltered lexicon were such concepts. Excluding them leaves the themes of Section 3.4 unchanged. Other weaknesses remain: the per-tier curation thresholds equal the defaults, so no tier-specific calibration took effect; 168 cell-type prompts name marker genes that an image model cannot see; and every concept is scored with a single prompt.

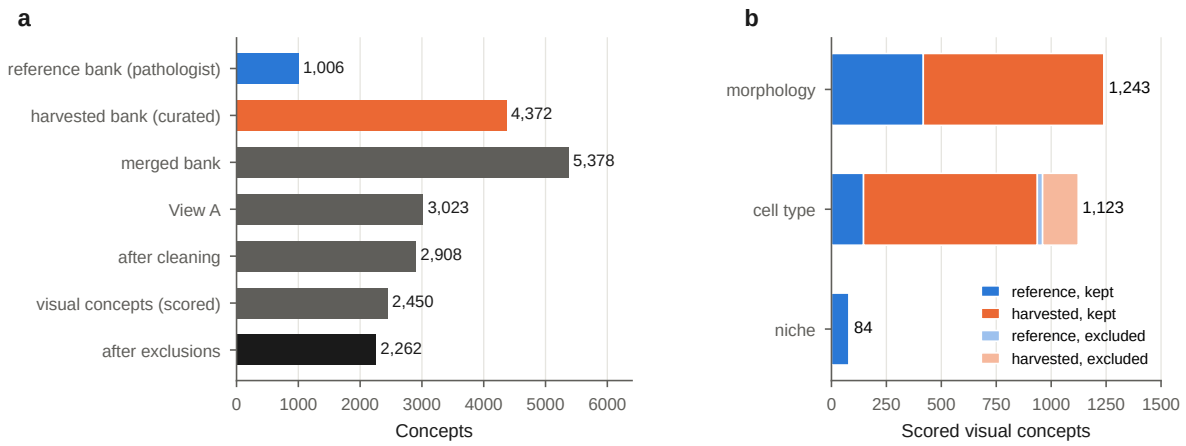


Figure 7. Concept vocabulary. **a**, Size of the vocabulary at each stage. View A comprises the 1,006 reference concepts, 255 matched harvested concepts and 1,762 harvested concepts close to the reference vocabulary. **b**, The scored visual concepts by tier and source, with the concepts excluded in the filtered analysis shown in light shades.

4 Discussion

The experts carry a technology signal. The clearest structure in the routing is not morphological. Expert 2 is used almost only on Xenium slides, each expert's concept profiles agree best between slides of the same data source, and the prediction error is dominated by the platform of the slide. The two platforms measure expression differently. Visium captures the whole transcriptome on 55 μm spots, whereas Xenium images the transcripts of a targeted gene panel in situ at subcellular resolution [7]. As a result, the targets differ in which genes are measured, in scale and in sparsity (Section 3.2). For the model this is a batch effect [12]: a systematic difference between groups of samples that is unrelated to the biology of interest. With targets this different, it makes sense for the router to separate the platforms, since they need different velocity fields. The images can give the router the cues it needs, since tissue preparation, staining and scanning also tend to differ between studies that used different platforms. In this data set platform is confounded with study and cancer type, so the analysis cannot say how much of the specialisation is technology, how much is cohort and how much is tissue.

What the alignment shows. The concept alignment describes what an expert's preferred patches look like in CONCH's embedding space. It does not show what the expert computes. It is correlational, depends on the vocabulary and on the wording of each prompt, and is weakest for the experts the model rarely uses. The low cross-slide stability means that a single pathology name per expert, the "clinical lexicon" envisaged at the start of the project, is not supported. The themes are better read as tendencies that hold on some slides and not on others.

Limitations. The test set has 23 slides, 7 of them Xenium, from a small number of studies, and some test slides share patients with training slides. The concept vocabulary was not reviewed by a pathologist beyond the reference bank, the filter removes invalid entries without fixing the curation, and disease names remain in the morphology tier. Two parts of the original plan were not carried out: the comparison of visual concepts with predicted Hallmark activity, and the ablation that replaces the Gaussian source of the flow by the image embedding ("direct flow matching").

Future work. Three directions follow from the results. First, the platform effect can be addressed directly, by conditioning the model on the platform as it is already conditioned on

cancer type, by harmonising the targets across platforms (for example, restricting the loss to genes measured on both), and by reporting results per platform. Second, the vocabulary needs a species filter for the Cell Ontology, a visibility review, calibrated thresholds and prompt ensembles, followed by re-embedding and a new expert analysis. Third, causal tests such as forcing or ablating an expert would show how routing changes the predicted expression and would turn the correlational lexicon into a functional one. The transcriptomic concepts (458 gene programmes and pathways) are already embedded and can support the planned comparison with predicted Hallmark activity.

5 Conclusion

Retraining MoLF on CONCH features preserves a working expression predictor and makes the experts accessible to text-based interpretation. The resulting expert lexicon is coherent on average (lymphoid, inflammatory, glandular and stromal themes) but unstable from slide to slide, agreeing mainly between slides of the same cohort, and both routing and error are organised as much by the measurement platform as by morphology. For models trained on several spatial transcriptomics platforms, the platform, and ideally the cohort, should be controlled for before any expert is given a biological name.

Code and data availability

The code, the frozen evaluation and interpretability outputs, and the concept vocabulary with its embeddings are available at https://github.com/rzadevv/MoLFDecoding. HEST-1k and the CONCH weights are available from their authors under non-commercial licences. The model checkpoints are not distributed; their SHA-256 hashes are recorded in the repository.

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